Riya Mahesh

New York University, Courant Institute of Mathematical Sciences
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EDUCATION

New York University, Courant Institute of Mathematical Sciences *Sep 2025 - Jun 2027*
Master of Science in Computer Science ***CGPA : 3.889/4.0*** *New York, USA*

Indian Institute of Technology Madras *Nov 2021 - July 2025*
Bachelor of Technology in Electrical Engineering, Minor in Computing ***CGPA : 3.95/4.0*** *Chennai, India*

Scholastic Achievements: 99.992 national percentile(**State Female Topper**) JEE Mains, KVPY & NTSE National Scholar
Relevant Courses: Computer Vision, Deep Learning for Imaging, Linear Optimization, Randomized Algorithms, Multi-Armed Bandits, NLP, Data Structures, Operating Systems, ML, Computer Organization, Control Systems, Digital Systems

SKILLS

C, C++, Python, Git, Tensorflow, PyTorch, HuggingFace, Matplotlib, Scikit, Numpy, Pandas, OpenCV, spaCy, CUDA, AWS

PROFESSIONAL EXPERIENCE

Morgan Stanley - Quant Software Development Intern *May 2024 - Jul 2024*
Feature Identification and Pattern Analysis for Data Comparison *Mumbai, India*

- Implemented **Credit & Bond Risk** pattern analysis algorithm, performing **feature attribution** of key difference drivers
- Generalized the tool with a custom UI for feature attribution of differences in any datasets, facilitating software migration, and received a **Pre-Placement Offer (PPO)** based on performance

Software and Embedded Engineer (Robotics) - Team Anveshak, IIT Madras *Aug 2022 - Feb 2023*
Autonomous Mars Rover with Drone *Chennai, India*

- Implemented **Aruco marker**, **object detection** using **OpenCV** & Path Planning Algorithms (A*, RRT) for rover traversal

Student Researcher - Indian Institute of Technology Madras *Aug 2023 - Jul 2024*
Temporal Causality Analysis for Climate Multivariate Time Series Data [GitHub] *Chennai, India*

- Engineered a **compute-efficient sparse attention pipeline** for **spatio-temporal Granger causality**, leveraging **top-k temporal pruning** and **masked spatial routing** to achieve **10% F1/ROC** over **VAR/XGBoost** methods

Indian Space Research Organisation (ISRO) - AI & ML Intern *May 2023 - Jun 2023*
Anomaly detection for satellite time series data using dynamic thresholding *Bangalore, India*

- Implemented **Outlier detection**, **Statistical** methods & **Autoencoders** for scoring anomalies in satellite **Time-series data**; modeled **LSTM** network to detect anomalies with Non parametric Dynamic Thresholding achieving **90%** accuracy

Research Assistant - Nanyang Technological University *May 2023 - Nov 2024*
Privacy and Security for Cloud-based AI & Distributed Training Systems [GitHub] *Singapore*

- Optimized **cross- node communication cost** in **CNN distributed training** via **Gossip Learning** on Sparse graphs of size 150 nodes, achieving 30.7 % lower cost with a **Minimum Spanning Tree**-based algorithm
- Surveyed cloud security guarantees & identified ML lifecycle stages prone to **Poisoning**, **Evasion**, **Exploratory** attacks and proposed a **taxonomy** for different threat models and **cryptographic defenses** prevalent in the current ML scenario

PUBLICATIONS

[1] Transformers with Sparse Attention for Granger Causality [Springer]
[2] Privacy and Security Implications of Cloud-Based AI Services: A Survey [arXiv]
[3] Efficient Model Propagation for Peer-to-Peer Federated Learning using MST and Gossip Networks [ACM FL-AsiaCCS'25]

PROJECTS

Zeroth Order Prefix Fine Tuning *Oct 2025 - Present*
Prof. Rajesh Ranganath, Courant Institute of Mathematical Sciences *New York University, Courant*

- Working on **memory efficient Zeroth order prefix fine-tuning** to improve In-context Learning in production LLMs

Information Retrieval System [GitHub] *Natural Language Processing*

- Built a custom search engine using **Vector Space Model** and **Bayesian Spell Check** & tested on Cranfield Data improving mAP, Recall, n-DCG metrics by **10%** by implementing **LSA**, **Word2Vec**, **Doc2Vec**, **BM25** algorithms

Anomaly Preserving Image Restoration for Noisy Images [GitHub] *Deep Learning for Imaging*

- Implemented custom **Residual Channel Attention Network**, **Denoising Autoencoders** and **U-Net** models to denoise images while preserving defects achieving highest average **PSNR** of **30.4** and **SSIM** of **99.5**

Virtual Memory Management and Multithreaded Process Scheduler [GitHub] *Operating Systems*

- Built a **page-based virtual memory subsystem** and **preemptive multithreaded process scheduler**, handling context switches, address translation, and concurrent execution simulating real-time OS operatiobs